

SAFE CAPSULE — GENERATION 1

100-Document Engineering Master List

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Purpose

Proposed controlled baseline for developing, analyzing, verifying and documenting a detachable occupant-survival capsule. The list is organized around traceability from requirements through architecture, safety analysis, engineering analysis, verification, testing and certification.

Status note

This is a proposed document architecture, not proof that the listed analyses have been completed. The reviewed source material identifies several performance values and mechanisms as conceptual and requiring CFD, FEA, simulation, testing and certification work.

A. Program, Requirements & Configuration Management

ID	Document	Purpose
SC-001	System Requirements Specification	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-002	Mission, Operational Concept & Use Cases	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-003	Design Objectives, Constraints & Design Philosophy	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-004	System Architecture Description	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-005	Product Breakdown Structure	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-006	Configuration Item List & Baseline	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-007	Requirements Traceability Matrix	Bidirectional trace from requirements to design and evidence.
SC-008	Interface Requirements Specification	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-009	Verification & Validation Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-010	Engineering Assumptions, Open Issues & Decision Log	Defines, analyzes or verifies the corresponding program or subsystem element.

B. Safety, Hazard Analysis & Human Protection

ID	Document	Purpose
SC-011	System Safety Program Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-012	Preliminary Hazard Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-013	Functional Hazard Assessment	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-014	Failure Modes, Effects & Criticality Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-015	Fault Tree Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-016	Common-Cause / Common-Mode Failure Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-017	Single-Point Failure & Loss-of-Safety Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-018	Emergency Scenarios & Safety State Definition	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-019	Occupant Protection Requirements	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-020	Safety Case / Safety Argument Structure	Defines, analyzes or verifies the corresponding program or subsystem element.

C. Capsule Geometry, Layout & Structural Architecture

ID	Document	Purpose
SC-021	Capsule General Arrangement	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-022	Cabin / Occupant Layout	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-023	Pressure Vessel / Cabin Boundary Definition	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-024	Primary Load Path Definition	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-025	Secondary Structure & Equipment Mounting	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-026	Material Selection & Allowables Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-027	Joints, Attachments & Fastener Design Basis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-028	Doors, Hatches, Windows & Emergency Exits	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-029	Center of Gravity & Mass Properties Report	Controls mass, CG and inertia inputs.
SC-030	Structural Design Criteria & Load Cases	Defines, analyzes or verifies the corresponding program or subsystem element.

D. Structural Analysis & Crashworthiness

ID	Document	Purpose
SC-031	Global Structural FEA	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-032	Local Stress / Joint FEA	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-033	Buckling & Stability Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-034	Fatigue & Damage-Tolerance Assessment	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-035	Dynamic / Transient Structural Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-036	Separation-Impulse Structural Load Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.

ID	Document	Purpose
SC-037	Landing Gear / Hard-Landing Load Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-038	Water-Impact / Slamming Load Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-039	Crashworthiness & Energy-Absorption Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-040	Structural Test & Instrumentation Plan	Defines, analyzes or verifies the corresponding program or subsystem element.

E. Separation, Egress & Release Mechanism

ID	Document	Purpose
SC-041	Separation System Architecture	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-042	Separation Interface Control Document	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-043	Separation Sequence & Timing Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-044	Separation Dynamics / 6-DOF Simulation	Quantifies relative motion, attitude and clearance after release.
SC-045	Separation Clearance & Collision Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-046	Release Mechanism Sizing & Load Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-047	Redundant Release / Initiation Logic	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-048	Separation Failure Modes & Recovery Logic	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-049	Ground Separation Test Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-050	Flight Separation Demonstration Plan	Defines, analyzes or verifies the corresponding program or subsystem element.

F. Descent, Parachute & Post-Separation Control

ID	Document	Purpose
SC-051	Descent System Requirements	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-052	Parachute Architecture & Deployment Concept	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-053	Parachute Sizing & Performance Analysis	Sizes descent system against defined environments and landing criteria.
SC-054	Canopy Deployment / Reefing Sequence Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-055	Parachute Failure & Partial-Deployment Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-056	Capsule Attitude-Control System	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-057	Post-Separation Propulsion / Control Sizing	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-058	Descent 6-DOF Simulation & Monte Carlo Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-059	Descent Test & Drop-Test Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-060	Landing / Splashdown Performance Verification Plan	Defines, analyzes or verifies the corresponding program or subsystem element.

G. Water Survival, Flotation & Marine Safety

ID	Document	Purpose
SC-061	Flotation Requirements & Buoyancy Budget	Closes buoyancy, stability and reserve requirements.
SC-062	Hydrostatic Stability Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-063	Trim, Freeboard & Damage-Flotation Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-064	Water-Impact Hydrodynamic Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-065	Capsize / Self-Righting Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-066	Hull / Keel Energy-Absorption Design	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-067	Watertight Integrity & Compartmentation	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-068	Sea-State / Wave-Environment Definition	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-069	Water Survival Equipment & Beacon Requirements	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-070	Model-Basin / Water-Test Program	Defines, analyzes or verifies the corresponding program or subsystem element.

H. Power, Electrical, Thermal & Life Support

ID	Document	Purpose
SC-071	Independent Capsule Power Architecture	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-072	Battery Sizing & Energy Budget	Closes independent capsule energy demand and reserve.
SC-073	Battery Thermal Runaway & Propagation Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-074	High-Voltage Isolation & Electrical Safety Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-075	Emergency Power Autonomy Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-076	Thermal Management Architecture	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-077	Fire Detection, Suppression & Smoke Management	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-078	Environmental Control & Life Support Requirements	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-079	Oxygen, CO2, Humidity & Cabin-Temperature Budget	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-080	Life-Support Verification & Emergency-Duration Test Plan	Defines, analyzes or verifies the corresponding program or subsystem element.

I. Avionics, Controls, Communications & Emergency Operations

ID	Document	Purpose
SC-081	Capsule Avionics Architecture	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-082	Independent Flight / Attitude Control Logic	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-083	Emergency Detection & Automatic Trigger Logic	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-084	Independent Navigation / Positioning Architecture	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-085	Communications, Distress Beacon & Tracking	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-086	Data Recording & Flight-Test Instrumentation	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-087	Human-Machine Interface & Crew Procedures	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-088	Cybersecurity / Secure Control Interfaces	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-089	Software & Hardware Assurance Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-090	Emergency Operations Manual & Checklists	Defines, analyzes or verifies the corresponding program or subsystem element.

J. Verification, Certification, Manufacturing & IP

ID	Document	Purpose
SC-091	Verification Compliance Matrix	Maps every requirement to an approved verification method.
SC-092	Requirements-to-Test Traceability Database	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-093	Certification / Regulatory Basis Assessment	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-094	Compliance Demonstration Strategy	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-095	Prototype & Development Test Article Definition	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-096	Manufacturing Process & Quality Assurance Plan	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-097	Inspection, Maintenance & Life-Limit Strategy	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-098	Reliability, Maintainability & Availability Analysis	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-099	Technology Readiness Level Assessment & Roadmap	Defines, analyzes or verifies the corresponding program or subsystem element.
SC-100	IP / Patent Architecture, Claims Map & Technical Evidence File	Defines, analyzes or verifies the corresponding program or subsystem element.

Document-Control Rules

- Every document should have an owner, revision, approval status, configuration baseline and change history.
- Every quantitative requirement should state units, source/basis, uncertainty where relevant, and verification method.
- Tag quantitative values as measured, calculated, assumed, allocated or target.
- Safety-critical functions require explicit failure detection, fault response and independent verification.
- Mass, CG, inertia, power and energy budgets should be controlled as living configuration documents.
- Simulation reports should state assumptions, boundary conditions, sensitivity studies and correlation needs.
- Test reports should preserve raw data, calibration, configuration, pass/fail criteria and deviations.
- Certification claims should not be presented as achieved until compliance evidence exists.
- The IP file should distinguish novel architecture from conventional elements and link claims to technical evidence.

Relationship to the reviewed material

The reviewed Gen-3 material already establishes a useful conceptual foundation: detachable capsule architecture, a 5–7 second target separation sequence, post-separation control, flotation and water-impact concepts, independent electrical architecture, risk/open-issue tracking and a verification matrix. This master list expands those themes into a controlled engineering-document structure without treating conceptual values as verified facts.